

Cross-coupling sweep — run report

RUN

233059_cross-sweep_phi2-sweep

Status: completed | Started: 2026-09-25T22:30:59.476556+00:00

PURPOSE

Sweep one V_{λ} at each fixed-axis bias across the other V_{λ} .

REQUIRED OPTICAL SETUP

PAX at final output; both paths open unless prompted.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port

Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

RUN COMMENT

same as last phi1-sweep but now with phi2. This is the second run on phi2.

OPERATOR SETUP NOTES

V_{π} expectation only; verify channel and measured driver gain before calibration.

DATA

1,716 recorded samples; acquisition: 1,716

Source: data.csv; full acquisition settings: recipe.json; status: run.json

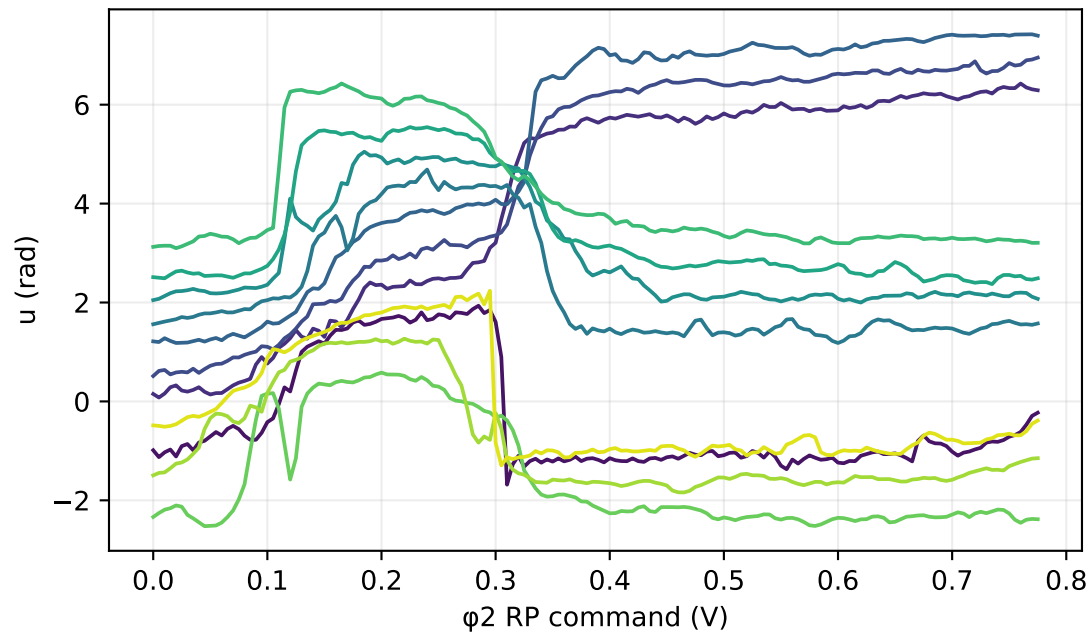
RAW DOP

Mean 0.9324; range 0.7768 to 1.0069. 2 samples outside 0–1. Telemetry is not a new calibration.

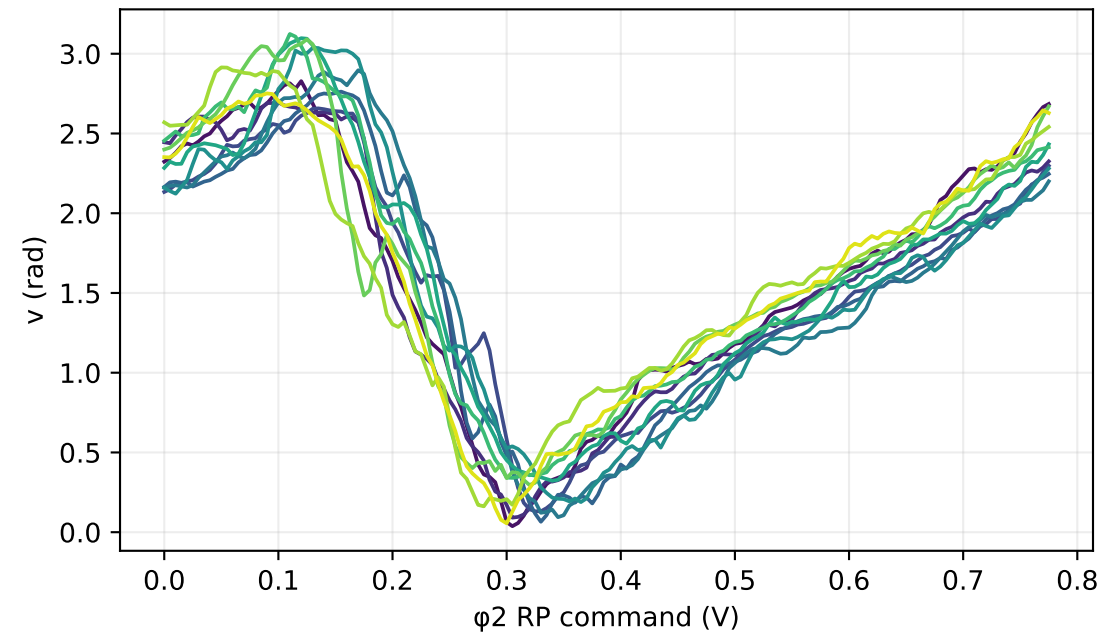
Cross-sweep bias relation — 233059_cross-sweep_phi2-sweep

swept ϕ_2 ; fixed ϕ_1 bias

Azimuthal response (unwrapped)



Polar response

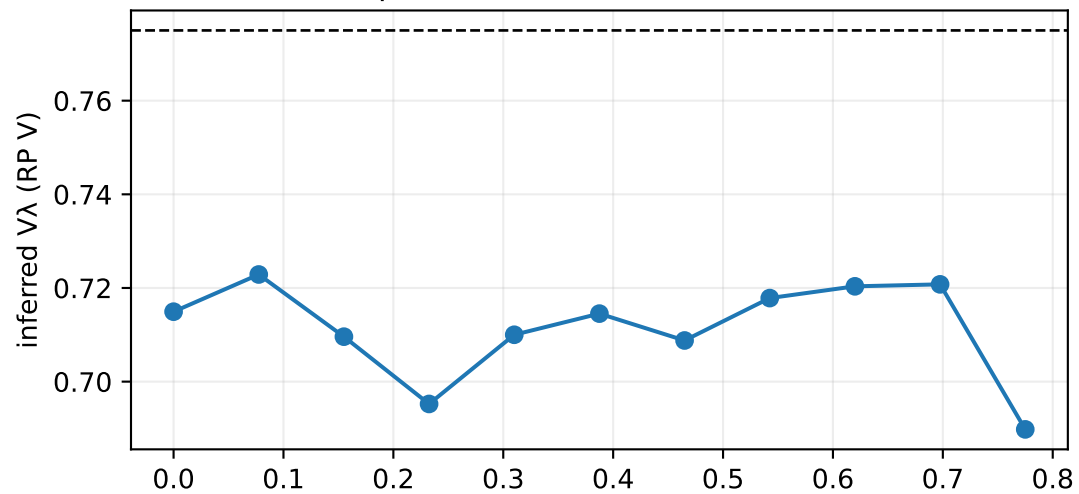


Recorded fixed bias

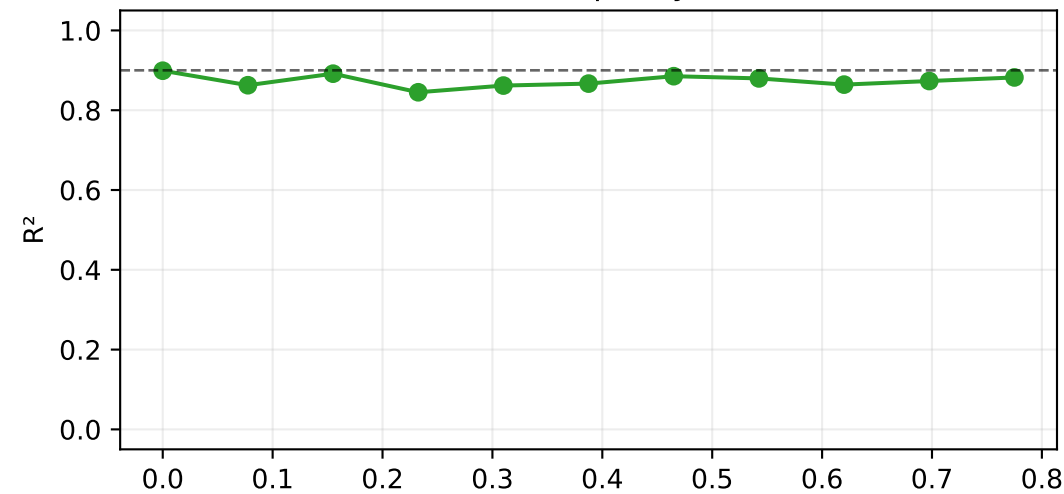


Cross-coupling summary — 233059_cross-sweep_phi2-sweep — swept ϕ_2 , biased ϕ_1

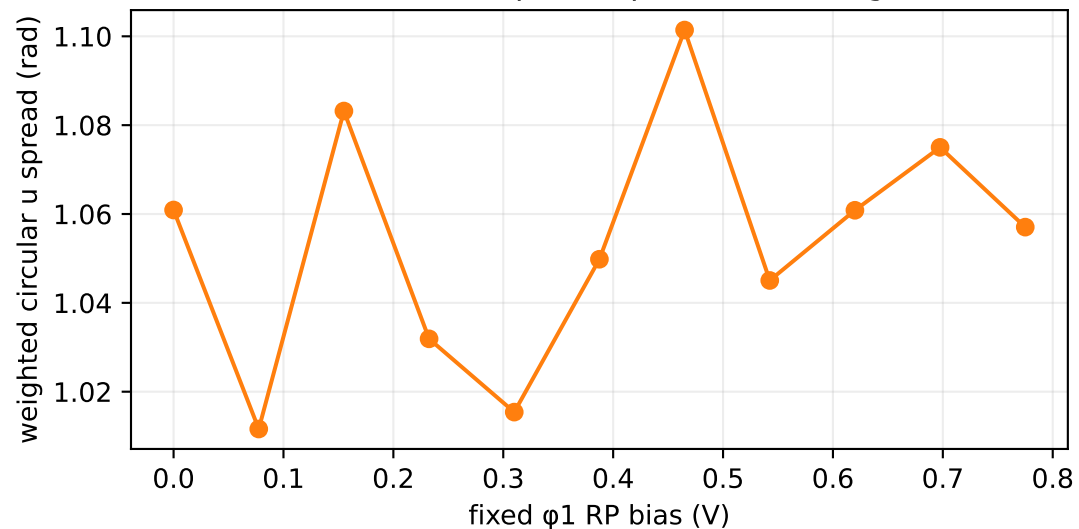
Principal-axis $V\lambda$ versus fixed-axis bias



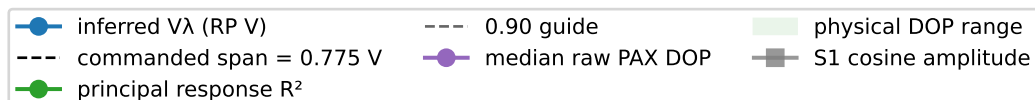
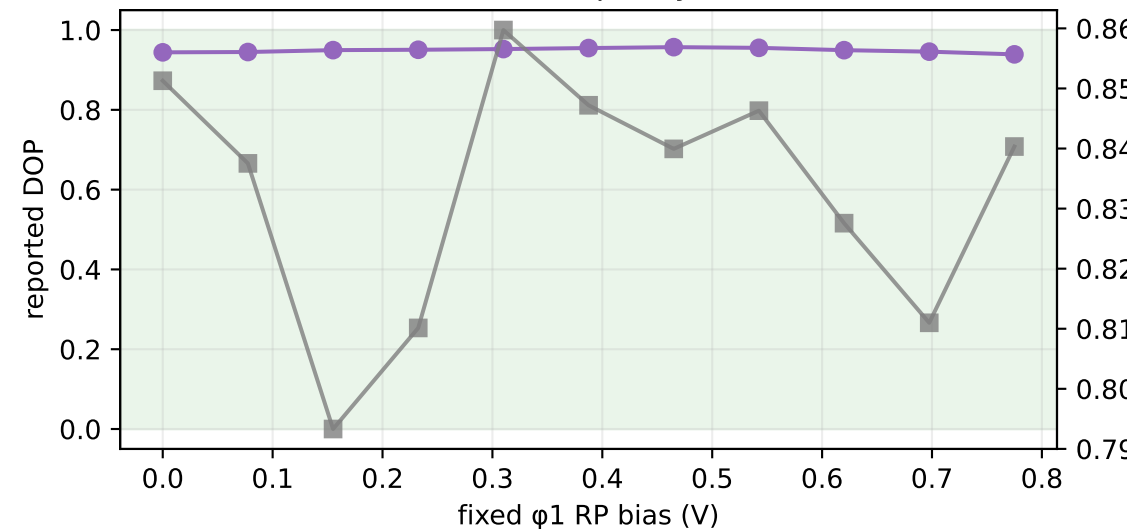
$V\lambda$ -fit quality



Off-axis azimuth spread (poles downweighted)



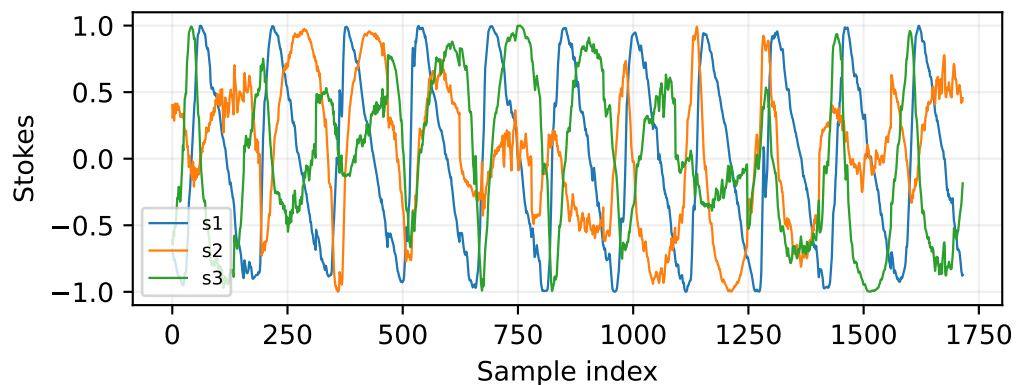
Measurement-quality context



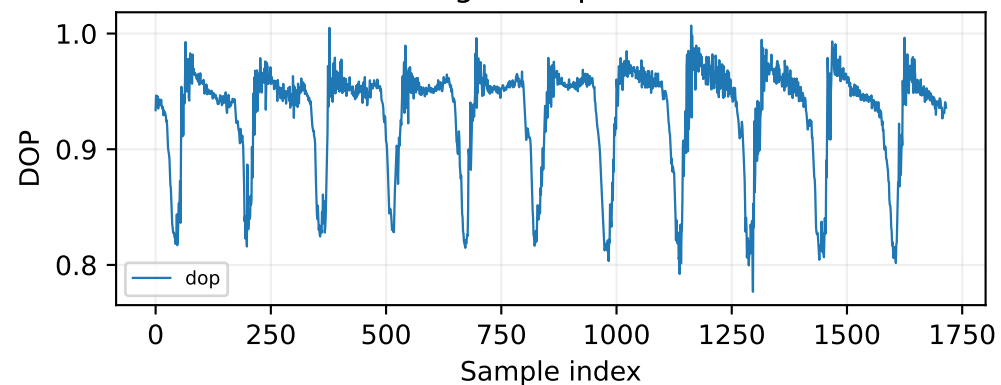
Recorded telemetry

Acquisition order preserves stage changes; no timing alignment or transfer gain is inferred.

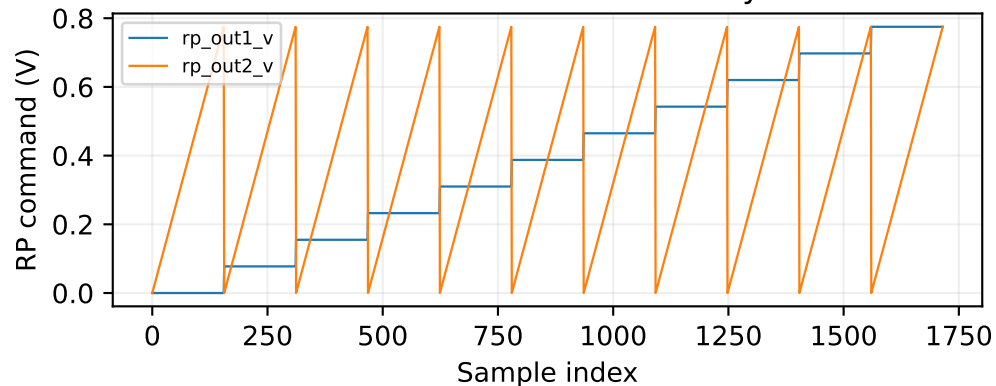
Normalized Stokes



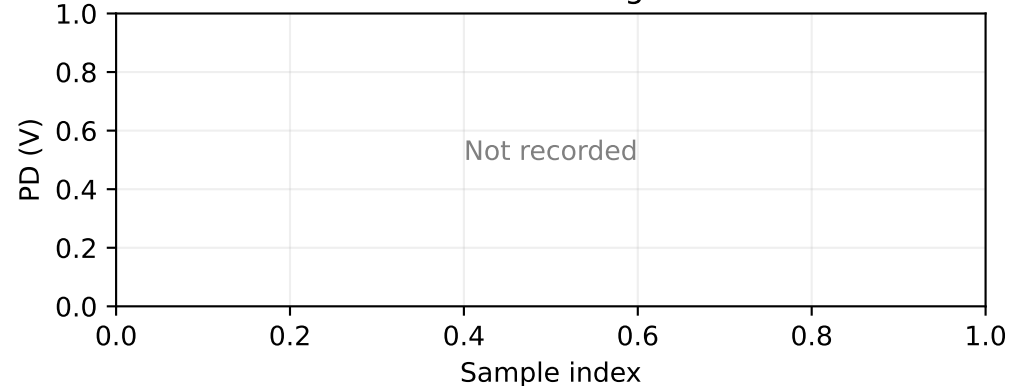
Raw degree of polarization



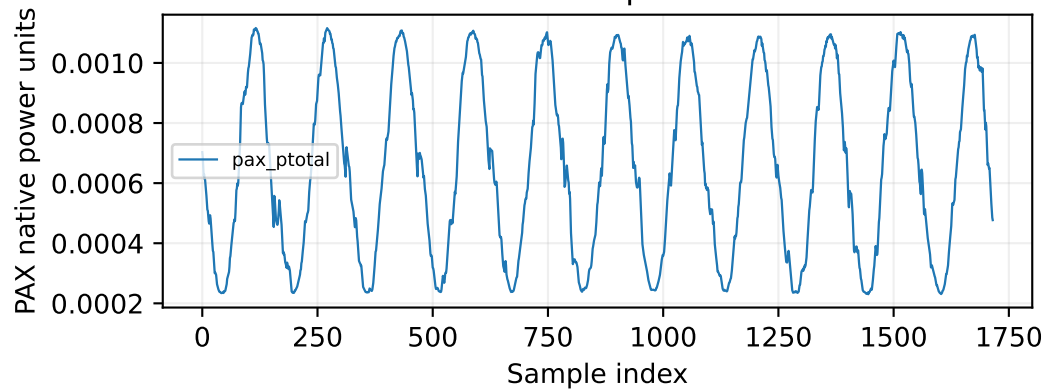
Actuator command history



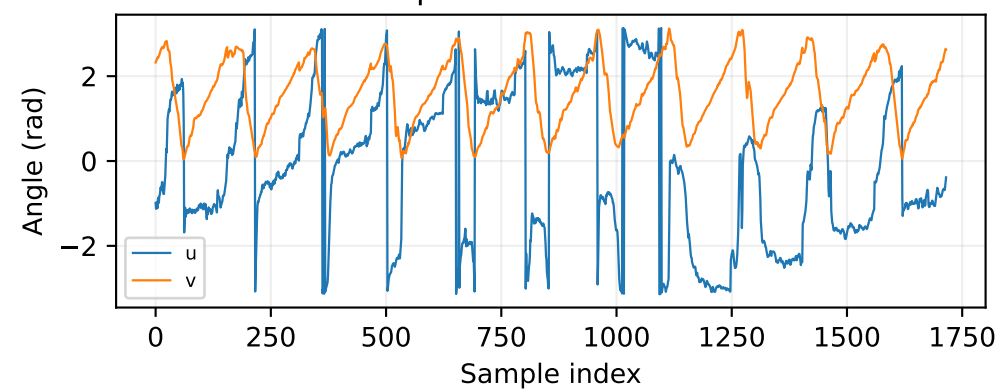
Photodiode signal



PAX total power



Sphere coordinates



Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π. Gains: terminal V / RP V.

Parameter	Recorded value
label	"phi2-sweep"
comment	"same as last phi1-sweep but now with phi2. This is the second"
axis	"phi2"
report	"pdf"
pax_rotation_velocity_hz	60.0
pax_measurement_mode	5
pax_wavelength_nm	830.0
phi1_v_lambda	13.078125
phi2_v_lambda	13.078125
phi1_actuator_volts_per_rp_volt	16.875
phi2_actuator_volts_per_rp_volt	16.875
rp_output_min_voltage	0.0
rp_output_max_voltage	1.0
cross_sweep_bias_intervals	10
cross_sweep_step_voltage	0.005
cross_sweep_settle_s	0.25